

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5

Total Marks:50

Annexure A SENARIO BASED QUESTIONS

<i>Criteria</i>	Understanding of Scenario (2)	Identification of Key Issues (2)	Application of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I Tejaswini.M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Outlier Detection and Data Reduction in E-Commerce Analytics (L4, L5)

An e-commerce platform wants to predict customer purchasing behavior. Some customers make unusually large purchases, and several variables contain overlapping information.

Sample Dataset

Customer_ID	Age	Annual_Income (₹K)	Orders	Total_Spend (₹K)	Avg_Order_Value (₹K)	Visits
E1	24	40	10	80	8	25
E2	29	55	14	120	8.6	30
E3	35	70	18	170	9.4	40
E4	41	90	22	220	10	48
E5	47	110	25	280	11.2	55
E6	52	140	35	400	11.4	70
E7	60	200	50	900	18	100
E8	32	65	16	150	9.4	35

Tasks

A. Outlier Detection

i. Calculate the mean and standard deviation of Total_Spend.

Total_Spend (₹K)	80	120	170	220	280	400	900	150
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- Mean = 290
- Standard Deviation ≈ 249.05

ii. Detect outliers using the Z-score method.

$$Z = \frac{x - \mu}{\sigma}$$

Customer	Total Spend	Z-score
E1	80	-0.84
E2	120	-0.68
E3	170	-0.48
E4	220	-0.28
E5	280	-0.04
E6	400	0.44
E7	900	2.45
E8	150	-0.56

iii. Identify the customer with unusually high spending.

Customer **E7** has the highest spending (₹900K). It is an unusual observation and is treated as a potential outlier.

B. Feature Redundancy

i. Identify attributes that may be mathematically or conceptually related.

Related attributes

- Orders
- Total_Spend
- Avg_Order_Value

ii. Explain the redundancy between Orders, Total_Spend, and Avg_Order_Value.

Relationship:

$$\text{Total Spend} = \text{Orders} \times \text{Avg Order Value}$$

Therefore, these variables carry overlapping information.

iii. Select features that should be retained for modeling.

Features to retain

- Age
- Annual Income
- Orders
- Visits

Drop either **Total_Spend** or **Avg_Order_Value** to reduce redundancy.

C. Data Reduction

i. Explain how feature selection reduces dimensionality.

Feature Selection

Feature selection removes unnecessary or correlated variables, reducing dimensionality and improving model efficiency.

ii. Suggest whether Principal Component Analysis (PCA) could be used.

Can PCA be used?

Yes. PCA can combine correlated variables into principal components.

iii. Explain one advantage and one disadvantage of PCA.

Advantages

- Reduces dimensionality.
- Removes multicollinearity.

Disadvantages

- Principal components are difficult to interpret.
- Original feature meaning is lost.

Decision

Use **feature selection first** because redundant attributes are already known.

Q2. Normalization of Financial Data for Loan Risk Prediction (L3, L4, L5)

A financial institution uses income, loan amount, savings, and credit score to classify customers into different risk categories. Since the variables have different units and scales, normalization is required.

Sample Dataset

Customer_ID	Income (₹K)	Loan (₹K)	Savings (₹K)	Credit_Score
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L1	30	100	10	620
L2	45	150	18	650
L3	60	200	25	680
L4	80	250	35	720
L5	100	300	50	760
L6	140	400	70	800

Tasks

A. Min-Max Normalization

i. Normalize Income to [0,1].

Formula

$$\frac{x - \min}{\max - \min}$$

Income

Min =30, Max =140

Customer	Income	Normalized
L1	30	0.000
L2	45	0.136
L3	60	0.273
L4	80	0.455
L5	100	0.636
L6	140	1.000

ii. Normalize Credit_Score to [0,1].

Credit Score

Min=620

Max=800

Customer	Score	Normalized
L1	620	0.000
L2	650	0.167
L3	680	0.333
L4	720	0.556
L5	760	0.778
L6	800	1.000

iii. Explain why the transformed attributes can now be compared on a common scale.

Reason

All attributes now lie between **0 and 1**, allowing fair comparison.

B. Z-Score Normalization

i. Compute the mean and standard deviation of Loan.

Mean Loan

233.33

Standard Deviation

98.60

ii. Calculate the Z-score for each loan amount.

Loan	Z-score
100	-1.35
150	-0.85
200	-0.34
250	0.17
300	0.68
400	1.69

C. Comparison

i. Compare Min-Max and Z-score normalization.

Min-Max	Z-score
Range 0–1	Mean =0, SD=1
Easy interpretation	Suitable for statistical analysis
Sensitive to outliers	Less affected than Min-Max

ii. Identify which method is more affected by extreme observations.

Most affected by extreme values

→ Min-Max normalization

iii. Recommend a normalization technique for a distance-based classifier.

Recommendation

For distance-based classifiers (KNN, K-Means), use Min-Max normalization.

Q3. Data Discretization for Hospital Patient Risk Analysis (L3, L4, L5)

A hospital wants to classify patients according to age, blood pressure, and cholesterol level. Continuous measurements need to be converted into meaningful categories.

Sample Dataset

Patient_ID	Age	Blood_Pressure	Cholesterol	Risk
H1	22	110	160	Low
H2	28	118	170	Low
H3	35	125	185	Low
H4	42	135	200	Medium
H5	50	145	220	Medium
H6	58	155	240	High
H7	65	165	260	High
H8	72	180	290	High
H9	80	190	310	High

Tasks

A. Equal-Width Binning

i. Divide Age into **3 equal-width bins**.

Age:

22–80

Range = 58

Number of bins = 3

ii. Calculate the bin width.

Bin width

$$58/3 \approx 19.3$$

iii. Assign each patient to a bin.

Bins

Bin1 (22–41.3)

H1, H2, H3

Bin2 (41.3–60.6)

H4, H5, H6

Bin3 (60.6–80)

H7, H8, H9

B. Equal-Frequency Binning

i. Divide Cholesterol into **3 approximately equal-frequency bins**.

Cholesterol values

160,170,185,200,220,240,260,290,310

ii. Assign patients to the corresponding bins.

Three bins

- Bin1 → 160,170,185
- Bin2 → 200,220,240
- Bin3 → 260,290,310

iii. Compare the results with equal-width binning.

Unlike equal-width binning, each bin contains nearly the same number of patients.

C. Concept Hierarchy

Construct the following hierarchy:

Age → Young / Middle-aged / Senior

Construct another hierarchy:

Cholesterol → Normal / Moderate / High

Age

- Young → 22–35
- Middle-aged → 36–59
- Senior → 60+

Cholesterol

- Normal → ≤ 180
- Moderate → 181–239
- High → ≥ 240

D. Analysis

Discuss how discretization can improve the interpretability of patient-risk prediction results.

Discretization

- simplifies continuous values,
- improves interpretability,
- speeds up mining,
- makes medical decisions easier.

Q4. Data Transformation and Skewness Analysis in Online Advertising (L4, L5)

An online advertising company records the number of seconds users spend viewing advertisements. Most users stay for a short duration, while a few users stay for several minutes.

Sample Dataset

User_ID	Ad_View_Time (seconds)
U1	5
U2	8
U3	10
U4	15
U5	20
U6	30
U7	45
U8	60
U9	120
U10	300

Tasks

A. Equal-Frequency Binning

i. Divide the data into **3 bins** with approximately equal numbers of observations.

Data

5,8,10,15,20,30,45,60,120,300

ii. Assign each value to a bin.

Three bins

- Bin1 → 5,8,10
- Bin2 → 15,20,30
- Bin3 → 45,60,120,300

iii. Explain how the method handles short and long viewing times.

Short viewing times are grouped together while long viewing times remain in the last bin.

B. Z-Score Normalization

i. Calculate the mean and standard deviation.

Mean

$$\mu = \frac{5 + 8 + 10 + 15 + 20 + 30 + 45 + 60 + 120 + 300}{10} = \frac{613}{10} = 61.3$$

Standard Deviation

$$\sigma \approx 86.12$$

ii. Compute the Z-score for all observations.

User	Time (s)	Z-score
U1	5	-0.65
U2	8	-0.62
U3	10	-0.60
U4	15	-0.54
U5	20	-0.48
U6	30	-0.36
U7	45	-0.19
U8	60	-0.02
U9	120	0.68
U10	300	2.77

iii. Identify the observation having the largest positive Z-score.

The highest Z-score is:

- User U10
- Viewing Time = 300 seconds
- Z-score = 2.77

Therefore, U10 is the most extreme observation.

C. Min-Max Normalization

i. Transform all values into the range [0,1].

Formula

$$x' = \frac{x - \min}{\max - \min}$$

where

- Minimum = **5**
- Maximum = **300**

Range = **295**

ii. Calculate the transformed values.

User	Original Value	Normalized Value
U1	5	0.000
U2	8	0.010
U3	10	0.017
U4	15	0.034
U5	20	0.051
U6	30	0.085
U7	45	0.136
U8	60	0.186
U9	120	0.390

User	Original Value	Normalized Value
U10	300	1.000

iii. Explain how the maximum observation affects Min-Max normalization.

- The maximum value (300) is converted to 1.
- The minimum value (5) becomes 0.
- Because 300 is much larger than the other values, the remaining normalized values become compressed near 0.
- Hence, Min-Max normalization is sensitive to extreme values (outliers).

D. Log Transformation

i. Apply a logarithmic transformation to the original values.

Original Value	log10(Value)
5	0.699
8	0.903
10	1.000
15	1.176
20	1.301
30	1.477
45	1.653
60	1.778
120	2.079
300	2.477

ii. Explain why log transformation is useful for highly skewed data.

- It reduces positive skewness.
- It compresses very large values more than small values.
- It reduces the influence of extreme observations.
- It produces a distribution that is closer to normal.

iii. Compare the transformed values with the original distribution.

Original Data	Log-Transformed Data
Highly skewed	Less skewed
Large gaps between values	Values become closer together
Outliers dominate the data	Outlier effect is reduced
Difficult to analyze	Easier to analyze and model

Q5. Comprehensive Data Preprocessing for Telecom Customer Analytics (L3, L4, L5)

A telecom company is developing a customer-churn prediction system. The customer data contains missing values, outliers, inconsistent categories, and attributes measured on different scales.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Bill (₹)	Call_Duration (min)	Data_Usage (GB)	Complaints	Churn
T1	22	M	450	120	5	1	No
T2	NaN	Male	600	180	8	2	No
T3	35	F	750	NaN	12	3	Yes
T4	48	Female	5000	900	50	15	Yes
T5	31	female	800	250	15	NaN	No
T6	42	FEMALE	950	300	18	4	Yes
T7	27	male	550	150	7	1	No
T8	65	Male	4500	700	45	12	Yes

Tasks

A. Missing Value Handling

i. Identify all missing values.

Attribute	Missing Value
Age	T2
Call_Duration	T3
Complaints	T5

ii. Apply median imputation to Age and Call_Duration.

Age

Available values:

22, 35, 48, 31, 42, 27, 65

Sorted values:

22, 27, 31, **35**, 42, 48, 65

Median Age = 35

Therefore,

T2 Age = 35

Call Duration

Available values:

120,180,900,250,300,150,700

Sorted values:

120,150,180,**250**,300,700,900

Median Call Duration = 250 minutes

Therefore,

T3 Call Duration = 250 minutes

iii. Apply an appropriate imputation technique to Complaints.

Available values

1,2,3,15,4,1,12

Sorted

1,1,2,**3**,4,12,15

Median Complaints = 3

Therefore,

T5 Complaints = 3

iv. Justify the selected methods.

- Median imputation is less affected by extreme values.
- It preserves the central tendency of the data.
- It is more suitable than mean when outliers are present.
- Hence, median imputation is appropriate for Age, Call Duration, and Complaints.

B. Outlier Detection

i. Detect outliers in Monthly_Bill and Call_Duration using Z-score.

Formula

$$Z = \frac{x - \mu}{\sigma}$$

Monthly Bill

Values

450,600,750,5000,800,950,550,4500

Mean

$$\mu = \frac{13600}{8} = 1700$$

Standard deviation

$$\sigma \approx 1817$$

Customer	Monthly Bill	Approx. Z-score
T1	450	-0.69
T2	600	-0.61
T3	750	-0.52
T4	5000	1.82
T5	800	-0.50
T6	950	-0.41
T7	550	-0.63
T8	4500	1.54

The **highest values (T4 and T8)** are potential outliers.

Call Duration

Values

120,180,250,900,250,300,150,700

Mean

$$356.25$$

Standard deviation

≈ 276.7

Customer	Call Duration	Approx. Z-score
T1	120	-0.85
T2	180	-0.64
T3	250	-0.38
T4	900	1.97
T5	250	-0.38

Customer	Call Duration	Approx. Z-score
T6	300	-0.20
T7	150	-0.75
T8	700	1.24

Again, **T4 and T8** have unusually high call durations.

ii. Identify customers corresponding to extreme values.

The customers having extreme values are:

- **T4**
- **T8**

These customers have unusually high monthly bills and call durations.

iii. Apply capping or transformation to the identified outliers.

Two possible techniques are:

Capping (Winsorization)

Replace extremely large values with an upper acceptable limit.

Example

- 5000 → 4500
- 900 → 700

OR

Log Transformation

Apply

$$\log(x)$$

This compresses very large values while preserving their order.

C. Categorical Standardization

i. Standardize Gender into Male/Female.

Original	Standardized
M	Male
Male	Male
male	Male
F	Female
Female	Female
female	Female
FEMALE	Female

ii. Explain why inconsistent categorical labels may produce incorrect model results.

If "Male", "male", and "M" are treated as different categories:

- Duplicate categories are created.
- Incorrect frequencies are calculated.
- Classification accuracy decreases.
- The model becomes less reliable.

Therefore, standardization improves data consistency.

D. Data Transformation

i. Apply Min-Max normalization to Monthly_Bill.

Formula

$$x' = \frac{x - \min}{\max - \min}$$

Minimum = 450

Maximum = 5000

Range = 4550

Customer	Monthly Bill	Normalized
T1	450	0.000
T2	600	0.033
T3	750	0.066
T4	5000	1.000
T5	800	0.077
T6	950	0.110
T7	550	0.022
T8	4500	0.890

ii. Apply Z-score normalization to Data_Usage.

Values

5,8,12,50,15,18,7,45

Mean

20

Standard deviation

≈16.72

Customer	Data Usage	Z-score
T1	5	-0.90
T2	8	-0.72
T3	12	-0.48
T4	50	1.79
T5	15	-0.30
T6	18	-0.12
T7	7	-0.78
T8	45	1.50

iii. Compare the two normalized attributes.

Min-Max	Z-score
Range 0–1	Mean = 0, SD = 1
Preserves fixed range	Indicates distance from mean
Suitable for KNN	Suitable for statistical analysis
Sensitive to outliers	Better for normally distributed data

E. Feature Selection

i. Identify potentially redundant attributes.

Possible redundant attributes:

- Monthly Bill and Data Usage
- Monthly Bill and Call Duration

These attributes may be highly correlated.

ii. Select the most relevant features for churn prediction.

Selected attributes:

- Age
- Gender
- Monthly Bill
- Data Usage
- Complaints

Target variable:

Churn

iii. Justify the selected subset.

These features:

- represent customer behavior,
- influence customer churn,
- remove unnecessary redundancy,
- improve prediction accuracy,
- reduce computational complexity.

F. Overall Analysis

Prepare a preprocessing sequence for this dataset beginning with:

Data Cleaning → Missing Value Treatment → Outlier Detection → Standardization → Normalization → Feature Selection

Explain why this sequence is suitable for preparing the data for a classification model.

Step 1 – Data Cleaning

Remove duplicate records and identify missing values.

↓

Step 2 – Missing Value Treatment

Replace missing Age, Call Duration, and Complaints using median imputation.

↓

Step 3 – Outlier Detection

Identify extreme values using the Z-score method and apply capping or log transformation.

↓

Step 4 – Standardization

Convert Gender values into only **Male** and **Female**.

↓

Step 5 – Normalization

- Apply Min-Max normalization to Monthly Bill.
- Apply Z-score normalization to Data Usage.

↓

Step 6 – Feature Selection

Select only the important attributes required for churn prediction.

Why is this sequence suitable?

This sequence ensures that the dataset is **clean, complete, consistent, and properly scaled** before training a classification model. Missing values are handled first to avoid inaccurate calculations. Outliers are treated before normalization so that extreme values do not distort the scaled data. Categorical values are standardized to maintain consistency, and feature selection removes redundant attributes, improving model accuracy, reducing computational cost, and enabling better churn prediction.